

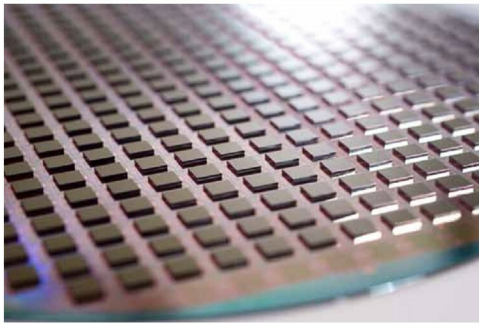
breakage, wafer misalignment, robot malfunctioning and even tool vibration due to environmental forces such as air flow. While these are general problems that have been haunting semiconductor fabs for years, newer yield-related problems have come about due to the reduced geometries. New techniques are emerging to conquer these problems.

New challenges. At 32 nm and below, the intricate process and layout interactions cause failures that are not foreseen by normal simulation tools. Irregularity in logic also causes failures. The very low 'k' factors in lithography are also causing problems these days. At the 32, 28 and 22nm levels, this leads to design sensitivities such as hot spots (areas on the die surface that exhibit excessive heating).

Modelling and simulation. Everything including system-level power, gate oxides, etch transfers and line-edge roughness has to be measured, modelled and simulated—nowadays even at the atomic level—to spot such problems. Some fabs monitor as many as 50,000 points. Of course, errors do not show up immediately by watching a single wafer. Sometimes, they are found only after monitoring hundreds or thousands of wafers.

From fab to design and back. The key goal behind such modelling and monitoring is to immediately identify problems and their root cause, and correct them at the design stage. All metrology data from the fab has to be fed back to the design centres, and design improvements flown back. This is what most latest diagnostic tools such as those from Mentor Graphics and Synopsys attempt to do. From lithography, manufacturing process to object-oriented process control, everything has to be taken care of at the design stage. This is leading to adoption of many techniques like design-based inspection and design-for-manufacturing (DFM).

"DFM is one area where improvements are being made specifically in phase-shift-mask and OLE process



Populated wafer

control (OPC) techniques. However, exactly how they help in improving the yield is not clear," says Prof. Jadcherla.

Identifying problems and improving the yield is no mean task and could take around a year to achieve even for the most experienced companies—as it happened during the introduction of the 40nm process at TSMC, the world's largest semiconductor manufacturer. When a new manufacturing process is introduced, it is first developed and refined in a test fab and then transferred to production lines in a process called 'chamber matching.' This theoretically ensures standard conformity and higher yields. Problems with chamber matching on TSMC's 40nm lines caused yield problems, thereby making it difficult for the company to match up to the demand for 40nm technology. It took almost a year for them to iron out these problems and plump up the yield of the process to match the previous generation's.

Improvements in lithography

Lithography is one area in which a lot of improvements have been made, and are expected in the near future—especially due to the advancements made

in photonics. As chips become smaller, there is an imminent move from 193 to 13.5nm-wavelength lithography in order to pattern finer features. This is due to a new lithography technology called 'extreme-ultraviolet' (EUV) (http://en.wikipedia.org/wiki/Extreme_ultraviolet_lithography) which could allow current chip features to shrink by almost ten times. Since 2009, a variety of EUV tools have been launched, including EUV scanners for process development and EUV light sources.

Apparently, EUV requires a completely new yet cost-effective and reliable light source. Versions have been demonstrated that hit 100W output power, enough for minimal production. However, high-volume manufacturing will require double that figure. A recently launched EUV source from Cymer Inc. of San Diego uses a high-powered infrared laser to bombard a microscopic molten tin droplet as much as 50,000 times per second. The resulting plasma radiates photons over a range of wavelengths. The 13.5nm light is collected and directed into the scanner illuminator. Japanese company Gigaphoton Inc. is also pursuing a similar laser-based approach. The